TagRepo - A tool to assign and query tags on files in a filesystem

BSc Computer Science

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Project Aims

Investigate and develop a tag-based file manager.

- Assign tags to files
- Search for files based on tags
- Should be easy to use for new users
  - UX is one of the focus of this project.

Digital file management is fundamental to computer usage.

- Required for workers or casual users
- Number of digital files is increasing - storage space increasing

Current systems are inefficient for managing a large number of files:

- Hierarchical structures (directory trees) - difficult to search for files (Henderson and Srinivasan, 2009) (Hicks *et al.*, 2008).
- Less flexible for organising and retrieving files (Henderson, 2011).

Project Objectives

Research:

- Identify challenges in PIM
- Evaluate existing tagging software
- Gain an understanding of PIM concepts
- Understand the needs of users when managing files
- Investigate challenges related to file management

Implementation:

- Develop a system - allow managing and searching files with tags
- Implement UI - easy to learn, and efficient

Evaluation:

- Evaluate effectiveness of system through user testing
- Evaluate system against existing software
- Come up with recommendations for further work

Programme of Study

A1: Must design, implement and test software to ensure it meets objectives - appropriate methodologies and tools *(e.g. design patterns, test-driven development)*

A2: Understand operating systems for software integration - small-scale implementation techniques *(e.g. dynamic programming, unit testing, benchmarking)*

A3: Understand the needs of users - appropriate usability principles and techniques *(e.g. participatory design, data collection)*

B2: Follow a plan, including timeline, milestones, objectives; use risk registers

B3: Project uses appropriate mathematics and engineering techniques - requires understanding of algorithms, data structures, software engineering principles

B4: Consider the ethical implications when developing the system *(e.g. user privacy and data security)*

B5: Final project report and presentation: should be clear and concise - requires understanding of technical details to present them effectively

C1: The project integrates various concepts from the field of computer science, such as operating systems, algorithms and user experience. This requires the use of appropriate techniques to ensure that the various components of the system are integrated correctly.

C2: The project investigates existing solutions to file management, as well as develops a new system to manage and search files using tags. This requires the use of critical thinking to evaluate the reliability and relevance of the knowledge acquired in order to solve the unfamiliar problems associated with the project.

C3: The project requires the communication of complex technical concepts, such as algorithms and data structures, to a variety of audiences, both through the production of a report and through the presentation of the project. This requires the use of appropriate communication techniques to ensure that the technical concepts are explained in an effective manner.

C4: The project requires the use of ethical principles, such as user privacy and data security, in order to ensure that the system is developed in a responsible manner.

C5: The project involves the development of a system to manage and search files using tags. This requires the use of creativity and innovation to develop a system which is different from existing solutions.

Why is this a Engineering/Computing problem?

We develop a tag-based file manager.

- Desktop application, interfaces with the underlying operating system
- Built using engineering principles and computing technologies
- Design system architecture such that it...
  - Allows assignment and quick searching of tags on files
  - Handle large numbers of files
- User interface should be easy to use, related to UX

Concepts in engineering and computing:

- Requirements gathering
  - Entity-relationship modelling
  - Use cases
- Backend language: Rust
  - Operating system (file operations, file tracking)
  - File systems
  - Object-oriented programming
  - Multi-threading, concurrency
  - Efficient programming (dynamic programming, recursion, complexity)

Concepts in engineering and computing:

- Database: SQLite
  - Database management systems
  - Database schemas
  - Database design
- Web technologies: JavaScript, Vue
  - Client-side programming
  - Web frameworks
  - HTML, CSS
  - UX, User Interface design
- Testing: Unit, integration, user
  - Automated testing
  - Performance testing
  - Usability testing

Literature Review Findings

Concepts

- Personal Information Management (PIM) - a field of research, concerned with how people store, organise, and retrieve information to complete tasks (Jones and Teevan, 2007).
- File management - the process of storing, retrieving and manipulating files on a computer system.
- Tagging - assigning labels to an item, enable users to search for files based on their content.

Previous approaches

- Directory trees
  - Used by most modern operating systems
  - File name duplication (Henderson, 2011)
  - Deep structures require users to make many navigational decisions (Hicks *et al.*, 2008)
- Tagging
  - Windows built-in tagging - doesn't support all filetypes, tagging interface is difficult to access.
  - TagSpaces - directly renames the file with a prefix containing the tags, causes issues with other software.
  - Tabbles - tags files using a backing database, but is not portable and locked to the system.

Lessons learnt

- Tagging may be more efficient than directory trees
- Many ways to implement tagging, depending on the use case.
- Additional features such as tag hierarchies, auto-tag rules to improve UX.
- Must support arbitrary filetypes, avoid modifying the existing file structure.

What is your proposed solution?

I will develop a tag-based file manager that allows users to assign tags to their files and search for them.

Structure / Implementation

- Rust backend
  - For file operations, database access and cross-platform compatibility.
  - Responsible for creating, reading, and updating files and tags.
- Vue frontend
  - For the graphical user interface and cross-platform compatibility.
  - Provides a user-friendly interface for tagging and searching files.
- SQLite database
  - For storing files and tags, and searching for files given a user query.

Functionality

- A folder can be designated as a "repository"
  - Similar to repositories in version-tracking software like Git
- The software is able to assign, search and manage tags for all files within a repository
  - Tags are not tied to the operating system, so the tags can be transferred between systems by moving the folder
- Keep track of files and tags using a database for each repository
  - Keeps existing file structure intact, avoid breaking compatibility with other software
- UI should be designed to be accessible and easy to use
- Provide a flexible and powerful search function

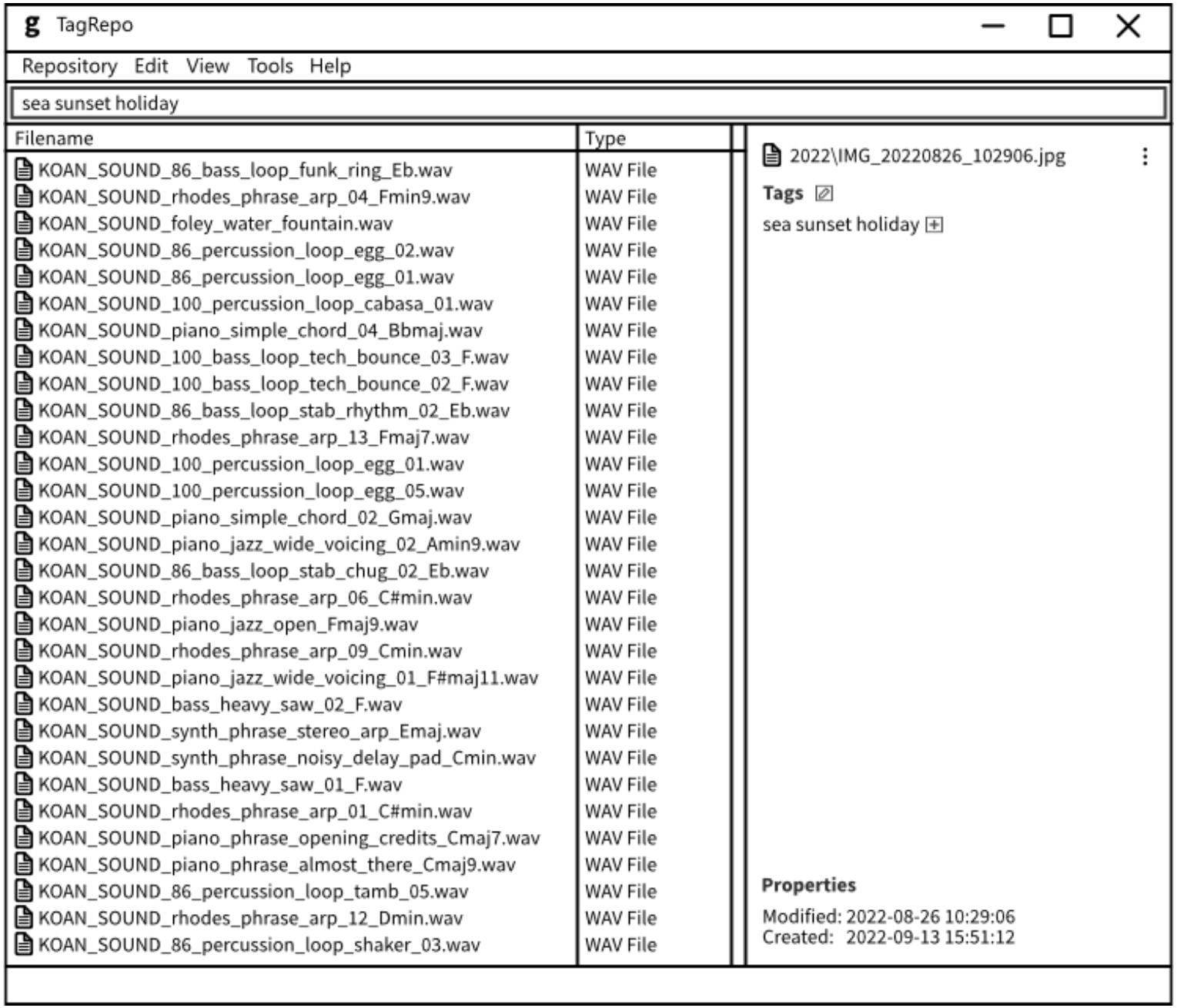

Figure 1: Mockups

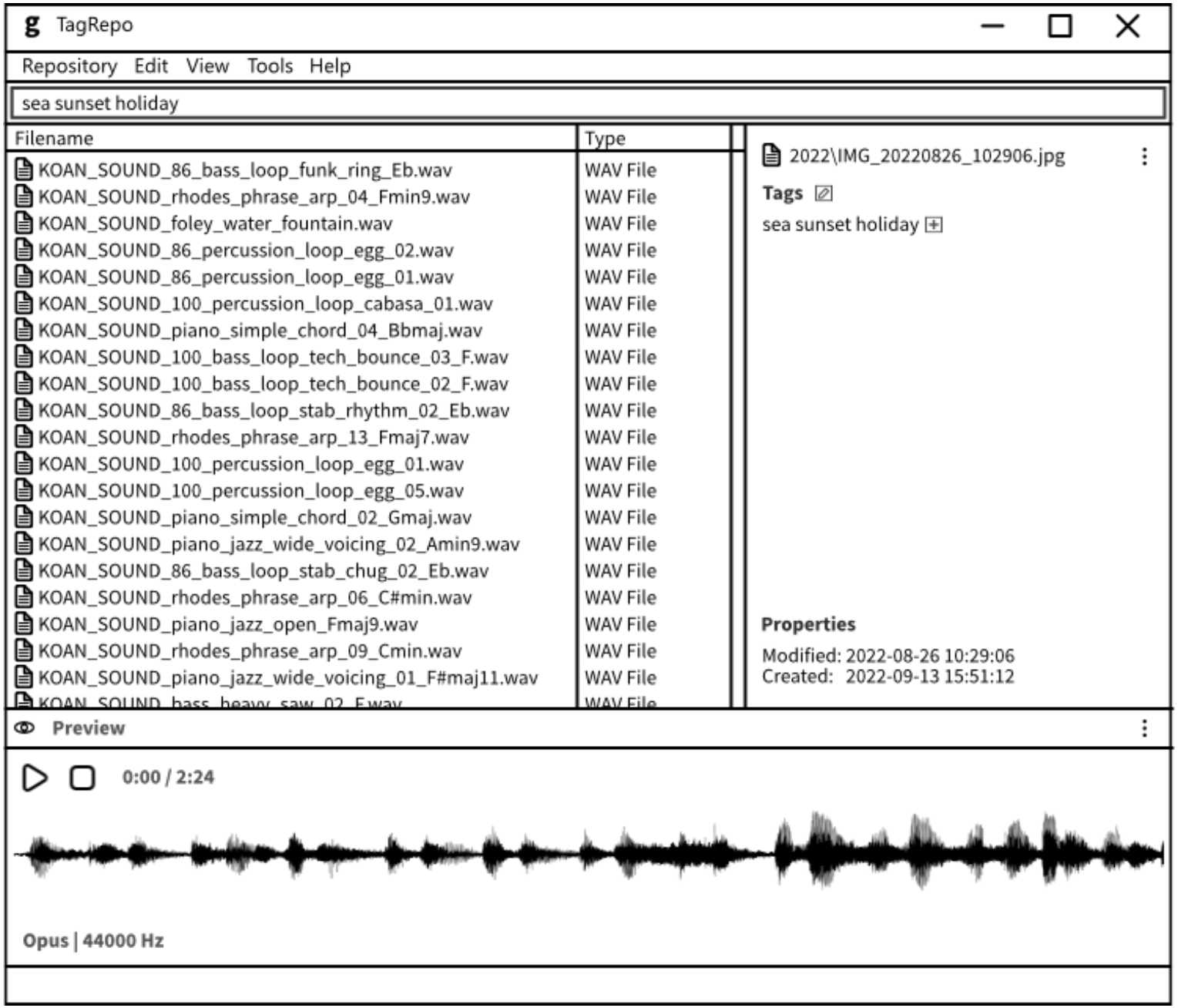

Figure 2: Mockups

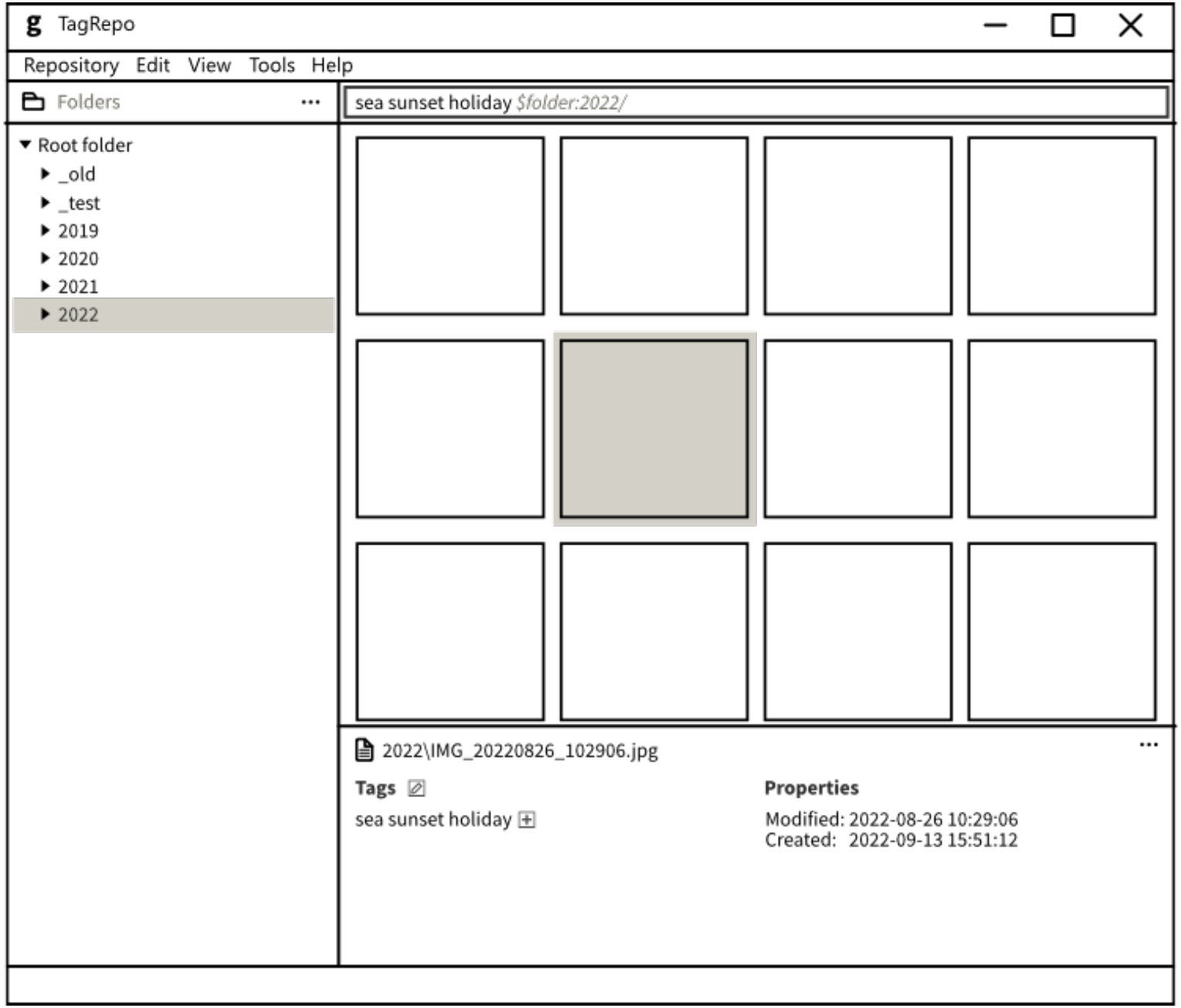

Figure 3: Mockups

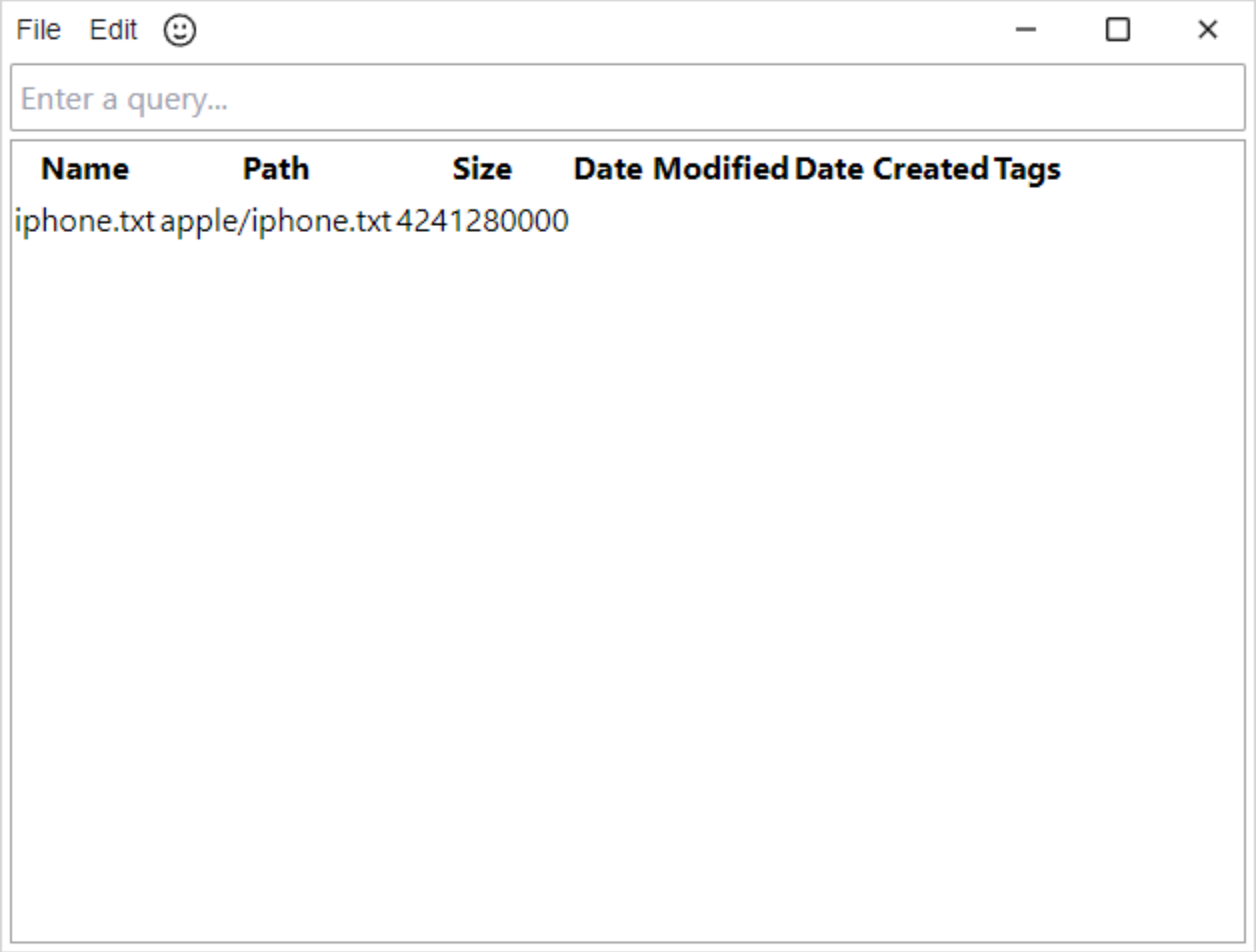

Figure 4: Current state

How will you evaluate your solution?

I will use 3 methods:

- Performed with university students, friends, etc:
  - Heuristic evaluation
  - Observational user study
- Performed online through forums, social media:
  - Questionnaires

Heuristic evaluation

- Finds usability problems with a software
- Performed using people who also know what heuristic evaluation is
  - e.g. Students studying in QMUL
- Advantages
  - Fast and cheap
  - Small number of people required (3-5 people)
  - Tends to find many problems
- Issues
  - Doesn't use actual users
  - Not task focused, only finds general interface problems

Observational user study

- Present users with a task to perform, and observe how they perform the task
- Performed using actual users
- Advantages
  - Finds usability problems related to specific tasks
  - Uses actual users
- Issues
  - More time consuming to set up

Questionnaires

- Distribute software to users, then ask for feedback using a form
- Performed using actual users
  - e.g. Online users
- Advantages
  - Cheap and easy to distribute (e.g. online Google forms)
  - You can build a picture based on respondent data
- Issues
  - Quality of responses depend on the questions you ask
  - The answers don't provide a lot of context

Project Planning

Done	Deadline	Milestone
X	October	Build any mini-project using Rust, Tauri and Vue to learn the framework and libraries
.	December	Initial implementation of database backend, using sqlite
.	February	More work on the backend implementation, early prototype of the application (focus on functionality rather than UI)
.	March	Polished prototype, minimal viable product
.	April-May	Project extensions and finalised software

Risk Register

Risk	Impact	Likelihood	Rating	Preventative actions
Cannot finish initial implementation by March	Cannot complete draft report with sufficient information	Low	High	Break down project into basic functionality and extensions, focus solely on basic functionality before moving onto extensions
Fail to find any users for testing	Cannot complete project evaluation	Low	High	Start finding users for testing as soon as initial implementation is complete

References

Henderson, S. (2011) 'Document duplication: How users (struggle to) manage file copies and versions', *Proceedings of the American Society for Information Science and Technology*, 48(1), pp. 1–10.
Henderson, S. and Srinivasan, A. (2009) 'An empirical analysis of personal digital document structures', in *Symposium on human interface*. Springer, pp. 394–403.
Hicks, B.J. *et al.* (2008) 'Organizing and managing personal electronic files: A mechanical engineer's perspective', *ACM Trans. Inf. Syst.*, 26(4), pp. 23:1–23:40. Available at: https://doi.org/10.1145/1402256.1402262.
Jones, W.P. and Teevan, J. (2007) *Personal information management*. University of Washington Press Seattle.